MIL-A-46545B (MU)
20 June 1966
SUPERSEDING
MIL-A-46545A (MU)
21 October 1963

MILITARY SPECIFICATION

ACTUATOR, EXPLOSIVE, LINEAR, M2 PARTS FOR

1. SCOPE

1.1 This specification covers the quality assurance provisions for the component parts for one type of electrically actuated, gas operated power source designated as Actuator, Explosive, Linear, M2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-P-10025 - Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1375

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DRAWINGS

ORDNANCE CORPS

8848559 - Plug and Wire Assembly
8848563 - Ferrule
8848556 - Bellows, Assembly

PUBLICATIONS

ORDNANCE CORPS

ET-8855752 - Equipment Tabulation Number

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Component parts.-The component parts shall comply with all requirements specified on Drawing (dwgs.) 8848556, 8848559 and 8848563 and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, distortion, unblended radii, surface defects, molding defects, dirt, grease, flux, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined. All packing components shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections

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set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time of the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

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4.2 Initial production inspection

4.2.1 Submission.-Prior to the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the Contracting Officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of 20 each of plug and wire, and bellows assemblies and 10 complete sets of parts (i.e., 10 each of every component) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. An initial production sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Components will be subjected by the Government to any or all of the examinations specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample, or portion thereof. Until an initial production

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sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used for defects other than critical, if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Ferrule (see dwg. 8848563).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			
Major: None defined			
Minor:	AQL 2.50 percent		
201.	Inside diameter	Gage	01001
202.	Outside diameter, maximum (max.)	Gage	01002
203.	Depth, max	Gage	01003
204.	Either chamfer missing	Visual	01004
205.	Evidence of poor workmanship (see 3.3)	Visual	01005

4.3.2.2 Plug and wire assembly (see dwg. 8848559).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Diameter under shoulder	Gage	02001
102.	Distance between bridge wires, max	Gage	02002
103.	Plug face not prepared	Visual	02003
Minor:	AQL 2.50 percent		
201.	Distance between lead wires, max	Gage	02004
202.	Length of insulated lead wires, max	Gage	02005
203.	Lead wire not tinned	Visual	02006
204.	Excessive flash on plug	Visual	02007
205.	Evidence of poor workmanship (see 3.3)	Visual	02008

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4.3.2.3 Bellows (see dwg. 8848564 covering a detail of dwg. 8848556).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Outside diameter, max.....	Gage	03001
102.	Outside diameter at closed end, max.....	Gage	03002
103.	Length of tube at open end, max.....	Gage	03003
104.	Inside diameter of tube, min.....	Gage	03004
Minor:	AQL 1.50 percent		
201.	Distance closed end to first flute.....	Gage	03005
202.	Total length, max.....	Gage	03006
203.	End not flat.....	Gage	03007

4.3.2.4 Bellows assembly (see dwg. 8848556).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Protrusion of jacket.....	Gage	04001
Minor:	AQL 1.50 percent		
201.	Distortion at open end.....	Gage	04002
202.	Jacket excessively above flush...	Visual	04003
203.	Unwiped solder on inside surface.....	Visual	04004
204.	Evidence of poor workmanship (see 3.3).....	Visual	04005

4.3.3 Testing

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4.3.3.1 Pull test (see dwg. 8848559) - Major defect - Code No. 02009.-This test shall be performed 100 percent. Any unit exhibiting slippage of one, or both, lead wires at either end of the plug shall be classed defective and removed from the lot. Testing shall be as specified in 4.4.1.

4.3.3.2 Insulation resistance (see dwg. 8848559) - Major defect - Code No. 02010.-This test shall be performed 100 percent after performance of the pull test. Any unit failing to meet the drawing requirement shall be classed defective and removed from the lot. Testing shall be as specified in 4.4.2.

4.3.3.3 Leakage (see dwg. 8848556) - Major defect - Code No. 04007.-This test shall be performed 100 percent. Any unit exhibiting leakage, as evidence by the presence of bubbles formed by air from within the unit, shall be classed defective and removed from the lot. Testing shall be as specified in 4.4.3.

4.3.4 Inspection Equipment.-Equipment Tabulation Number ET-8855752 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

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4.4 Test methods and procedures

4.4.1 Pull test.-This test shall be performed using the equipment specified in 4.3.4. After application of the required force to each lead wire, observation shall be made for slippage at the entry point of the wires into the plug and on the opposite face of the plug. Any unit exhibiting any slippage whatsoever, shall be classed defective.

4.4.2 Insulation resistance.-The plug and wire assembly shall be tested using the equipment specified in 4.3.4. After application of the required potential for the specified period, the resistance of the unit shall be observed. Any unit exhibiting a resistance less than that permitted shall be classed defective.

4.4.3 Leakage.-The bellows assembly shall be assembled to the test equipment specified in 4.3.4. After assembly, the unit shall be submerged in water to a depth of 2 inches, minimum. The specified air pressure shall be applied and observation performed for leakage, as evidenced by the presence of a stream of bubbles, of any frequency, from the unit. Any unit exhibiting leakage shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking

5.1.1 Level C.-The component parts shall be packed and marked in accordance with Specification MIL-P-10085.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

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6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (SMUPA Form 1010) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the EL by the following codes:

CDAF - Contractor's design responsibility on Army format in accordance with MIL-D-45608.
CDCF - Contractor's design responsibility on contractor format.
AD - Army design.
ADMU - Army design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

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6.3.4 Specification control drawing.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagramatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above.

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Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35mm microfilm process. Contractors shall submit oneccopy of the final design as a rolled set or flat set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 Submission of contractor designs.-Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the sate of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of Initial Production Sample.-Instructions, as to the location for the evaluation of the initial production sample, shall be obtained from the Contracting Officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NI.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project No. 1375-A1026

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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